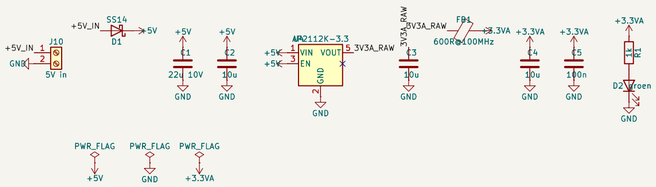
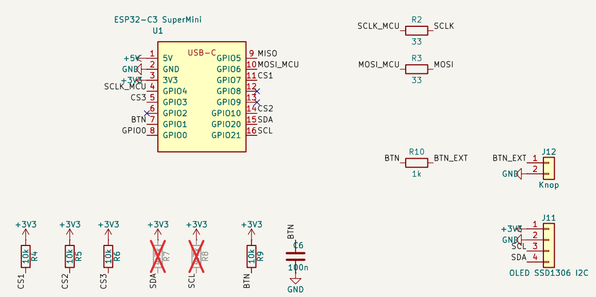
VOEDING (5 V via USB-kabel op J10)



+3.3VA = schone 3,3 V alleen voor de ATM90E32's

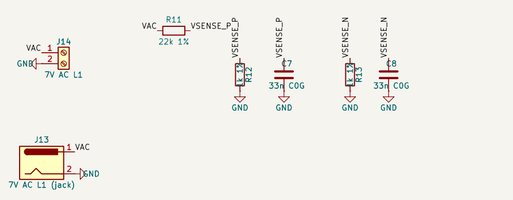
ESP32-C3 SuperMini (antenne naar bovenrand print)



J11: volgorde 3V3/GND/SCL/SDA — sluit aan met losse dupont-draadjes (OLED-pinvolgorde verschilt)

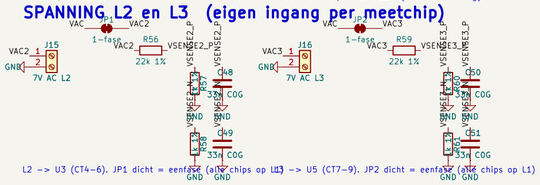
GPIO2/8/9 = strapping-pinnen, niet gebruiken

SPANNINGSMETING (7 V AC-adapter)



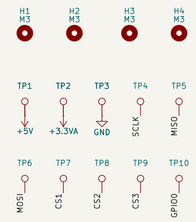
Deler 1/23: 10,5 V AC -> 457 mVrms. Geen serieweerstand op VSENSE_P (fase-matching)

SPANNING L2 en L3 (eigen ingang per meetchip)

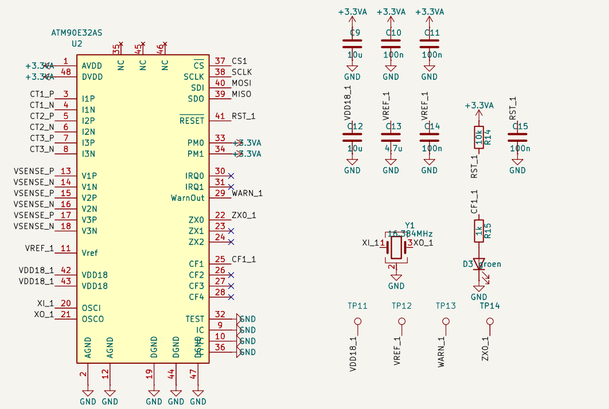


L2 -> U3 (CT4-6). JP1 dicht = eenfase (alle chips op U3). L3 -> U5 (CT7-9). JP2 dicht = eenfase (alle chips op L1)

MONTAGE + TESTPUNTEN

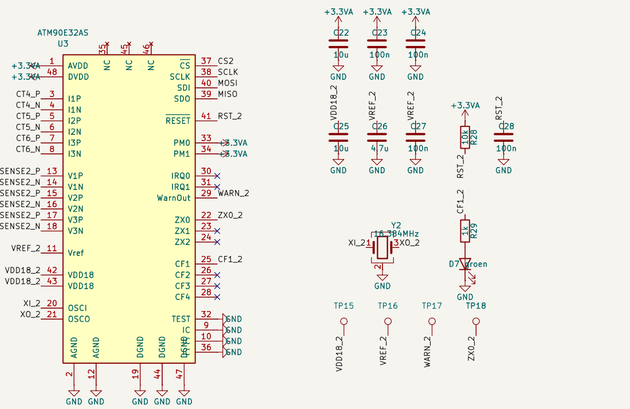


METEN 1: ATM90E32AS (CT1-CT3)



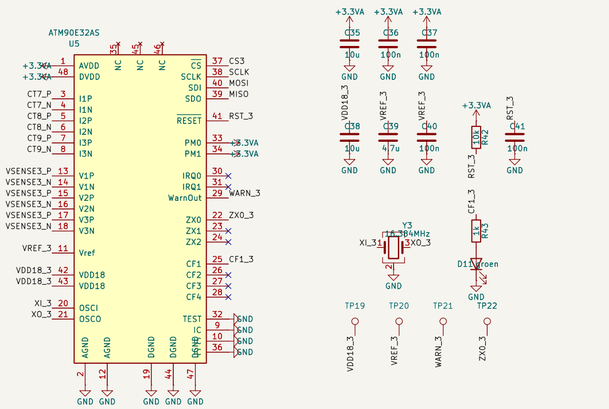
Kristal: geen load-C's (intern). PM0/PM1 hoog = Normal mode

METEN 2: ATM90E32AS (CT4-CT6)



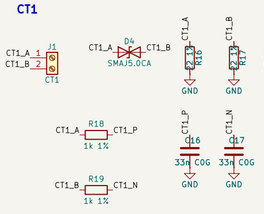
Kristal: geen load-C's (intern). PM0/PM1 hoog = Normal mode

METEN 3: ATM90E32AS (CT7-CT9)

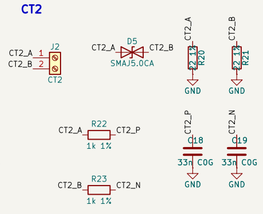


Kristal: geen load-C's (intern). PM0/PM1 hoog = Normal mode

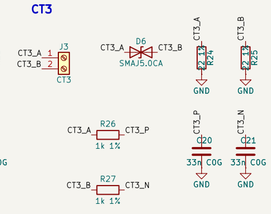
CT1



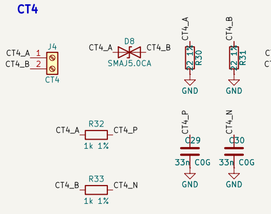
CT2



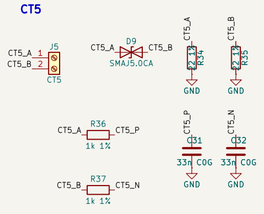
CT3



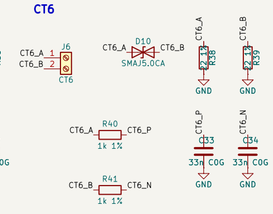
CT4



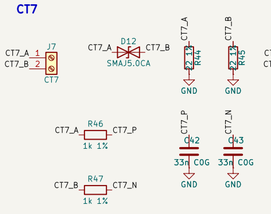
CT5



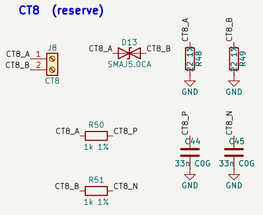
CT6



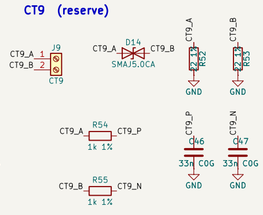
CT7



CT8 (reserve)



CT9 (reserve)



Burden 2x22R (44R) met middelpunt aan GND: 16 A -> 352 mVrms, max ~32 A. Filters 1k/33nF C0G (4,8 kHz).

CT's alleen aansluiten/loskoppelen met de groep UIT. Alle CT's dezelfde richting (pijl naar de belasting).

Gegenereerd startpunt: controleer elke ATM-pin tegen de datasheet
7 groepen + 2 reserve, eenfase. Zie Ontwerp_v2_ESP32-C3.md

Sheet: /		
File: PowerMeter.kicad_sch		
Title: PowerMeter v2 — ESP32-C3 SuperMini + 3x ATM90E32AS		
Size: A2	Date: 2026-11-06	**Rev: A**
KiCad E.D.A. 10.0.8		Id: 1/1